Algebra 1 – Module 3 Practice Name: _____

Relations and Functions • Sections 3.1 – 3.6

3.1 Representing Relations

1. Write the domain and range of {(−4, 2), (1, −3), (3, 2), (0, 6)}.

D = _____

R = _____

2. Write the relation as ordered pairs. Then give its domain and range.

x	−2	0	1	3
y	5	−1	5	2

3. "The number of hours a phone charges determines its battery percent." Name the independent and dependent variables.

3.2 Functions

4. Is {(5, 1), (−2, 1), (0, 3), (5, 8)} a function? Explain.

5. Is this mapping a function? Explain. 1 → 4, 2 → 4, 3 → 7

6. f(x) = 4x − 7. Find:

a) f(3)

b) f(−2)

c) f(0) − f(1)

7. Concert tickets cost t(x) = 15x dollars for x tickets. Find t(6) and explain what it means.

3.3 Linearity and Continuity

8. Linear or nonlinear?

a) $y = 6x - x + 2$

b) $y = 3x^3 - 1$

c) $2x + 7y = 14$

d) $y = 5/x + 1$

9. Discrete or continuous? Explain.

a) the number of tickets sold for a game

b) the height of a plant over a month

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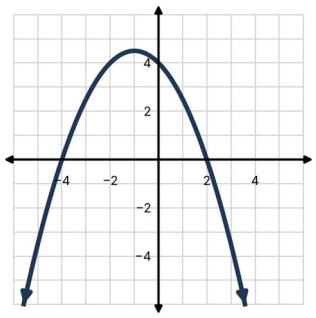
Relations and Functions • Sections 3.1 – 3.6

3.4 Intercepts of Graphs

10. Find the x- and y-intercepts of $3x - 6y = 18$.

11. A tank has $W = 480 - 32t$ gallons after t minutes. Find both intercepts and tell what each means.

Use the graph of f(x) for 12 – 13.

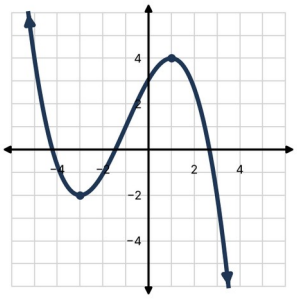


12. Name the intercepts. Where is f(x) positive? Negative?

13. Name the line of symmetry and describe the end behavior of f(x).

3.5 Shapes of Graphs

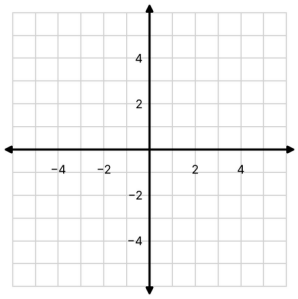
Use the graph of g(x) for 14.



14. Where is g(x) increasing? Decreasing? Name the relative maximum and minimum. Describe the end behavior.

3.6 Sketching Graphs

15. Sketch a graph with these key features: x-intercepts (−3, 0) and (3, 0); relative minimum (0, −4); symmetric about the y-axis; as x increases or decreases, y increases.



16. A hot-air balloon rises steadily for 20 minutes, floats at the same height for 30 minutes, then comes down steadily. Sketch height vs. time and label the increasing, constant, and decreasing parts.

